

Abelacimab in Cancer-Associated Thrombosis: The Right Drug at the Right Time for the Right Purpose. A Comprehensive Review.

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Cancer-associated thrombosis (CAT) is a devastating complication of cancer that can significantly impact a patient's health and life. The incidence of CAT is approximately 20%, and 1 in 5 cancer patients will develop CAT annually. Indeed, CAT can promote pulmonary embolism and deep vein thrombosis, leading to increased morbidity and mortality that dramatically impact survival. CAT can also provoke delay or discontinuation of anticancer treatment, which may result in a lack of treatment efficacy and high costs for patients, institutions, and society. Current guidelines advocate direct oral anticoagulants (DOACs) as the first-line anticoagulant option in CAT. Compared to low-molecular-weight-heparins (LMWHs), DOACs are advantageous in that they typically have an oral route of administration, do not require laboratory monitoring, and have a more predictable anticoagulant effect. However, in patients with thrombocytopenia, renal failure, or those receiving anticancer regimens with potential for drug-drug interactions, LMWH is still the mainstay of care. The main limitation of current anticoagulant agents is related to bleeding risk (BR), both for DOACs and LMWHs. Specifically, DOACs have been associated with high BR in gastrointestinal and genitourinary cancers. In this challenging scenario, abelacimab, an anti-factor XI agent, could represent a viable option in the management of CAT due to its "hemostasis sparing" effect. The safe profile of abelacimab could be useful in patients with active malignancy and CAT, as long-term anticoagulant therapy is often required. Two ongoing international phase III trials (Aster and Magnolia) compare abelacimab with the standard of care (i.e., apixaban in patients with CAT and dalteparin in those with CAT and high BR, respectively). Abelacimab is a new and attractive anticoagulant for the management of CAT, especially in the insidious and critical scenario of active cancer patients with venous thromboembolism and high BR. The aim of this narrative review is to discuss the updated evidence on the performance of DOACs and LMWHs in the treatment of CAT and to focus on the potential role of abelacimab in CAT and its promising associated clinical trials.

[Bempedoic acid and inclisiran : two new drugs to treat hypercholesterolemia].

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Last European Society of Cardiology (ESC) and European Atherosclerosis Society (EAS) guidelines on dyslipidemia published in 2019 recommend an LDL-cholesterol goal < 1.4 mmol/l (< 55 mg/dl) for people at very high cardiovascular risk. They propose combinations of lipid lowering therapies in addition to statins if necessary to meet this objective, which is very rarely reached in this at risk population. Two new drugs have recently received European Medicines Agency (EMA) approval for the treatment of hypercholesterolaemia, bempedoic acid, a small oral molecule that inhibits hepatic cholesterol synthesis, and inclisiran, a small interfering RNA that reduces PCSK9 levels after its subcutaneous injection. This article summarizes the mode of action, efficacy data, safety/tolerance profile and indications of these two innovative drugs.

[Bempedoic acid, new cholesterol-lowering drug].

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Patients at high or very high cardiovascular risk should have a LDL cholesterol level as low as possible, a target that may be difficult to be reached with statin monotherapy, especially when this treatment is not well tolerated. It is generally recommended to use pharmacological combinations to reach more and more strict objectives. Bempedoic acid is a new oral cholesterol-lowering medication that reduces hepatic cholesterol synthesis by inhibiting the enzyme adenosine triphosphate (ATP)-citrate lyase enzyme. Because this medication is a prodrug that is selectively activated in the liver, it is deprived of any muscular adverse effect. This article summarizes the mode of action, pharmacokinetics, efficacy, safety profile, indications and reimbursement conditions of bempedoic acid. This novel cholesterol-lowering drug is indicated in combination with statins (with or without ezetimibe) in patients who do not attain LDL goals despite administration of a statin at the maximum tolerated dose or in case of statin intolerance.

Bempedoic acid for the treatment of hypercholesterolemia.

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Although several lipid-lowering drugs are available, they are not sufficient for some patients. Bempedoic acid is a small molecule adenosine triphosphate-citrate lyase inhibitor indicated for the treatment of adults with hypercholesterolemia. We performed a systematic review of the literature using PubMed database, and the following keywords were used: 'bempedoic acid,' 'hypercholesterolemia,' and 'adenosine triphosphate citrate lyase.' The chemical property, mechanism of action, pharmacokinetics, clinical efficacy, and safety of bempedoic acid are introduced in this paper. Bempedoic acid can modulate the metabolism of cholesterol. Clinical trials indicated that bempedoic acid could significantly reduce low-density lipoprotein cholesterol levels. Bempedoic acid was well tolerated.

Bempedoic acid. Mechanism of action and pharmacokinetic and pharmacodynamic properties.

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Bempedoic acid acts by inhibiting adenosine triphosphate-citrate lyase (ACL) and consequently cholesterol biosynthesis, leading to increased expression of LDL receptors and increasing low-density lipoproteins (LDL-C) plasma clearance. It is a prodrug for oral administration with intracellular activation. It is activated in liver cells and to a lesser extent in kidney cells, being absent in adipose tissue and muscle cells. Therefore, unlike statins, its potential myotoxic effect is very limited. It has recently been approved as a lipid-lowering drug in combination with diet, with statins, or with other lipid-lowering drugs in patients with hypercholesterolaemia, mixed dyslipidaemia, statin intolerance, or when these are contraindicated. The marketing of bempedoic acid implies, in clinical practice, having a new family of lipid-lowering drugs.

Bempedoic Acid: for Whom and When.

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The aim of creating an orally active non-statin cholesterol-lowering drug was achieved with bempedoic

acid, a small linear molecule providing both a significant low-density lipoprotein cholesterol (LDL-C) reduction and an anti-inflammatory effect by decreasing high-sensitivity C-reactive protein. Bempedoic acid antagonizes ATP citrate-lyase, a cytosolic enzyme upstream of HMGCoA reductase which is the rate-limiting step of cholesterol biosynthesis. Bempedoic acid is a pro-drug converted to its active metabolite by very-long-chain acyl-CoA synthetase 1 which is present mostly in the liver and absent in skeletal muscles. This limits the risk of myalgia and myopathy. The remit of this review is to give clinical insights on the safety and efficacy of bempedoic acid and to understand for whom it should be prescribed. Bempedoic acid with a single daily dose (180 mg) reduces LDL-C by a mean 24.5% when given alone, by 18% when given on top of a major statin and by 38-40% when given in a fixed-dose combination with ezetimibe. Bempedoic acid does not lead to the risk of new-onset diabetes, and moderately improves the glycaemic profile. The extensive knowledge on bempedoic acid mechanism, metabolism and side effects has led to an improved understanding of the potential benefits of this agent and offers a possible alternative to cardiologists and clinical practitioners somewhat worn out today by the occurrence of the muscular side effects of statins.

[Bempedoic acid: mechanism of action].

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Bempedoic acid (ETC-1002) is an oral, once-daily, small molecule with a half-life of 15-24 h. It is responsible for the inhibition of ATP citrate lyase (ACLY), a cytosolic enzyme upstream of HMG-CoA reductase. Bempedoic acid is a prodrug rapidly converted in the liver to a coenzyme A derivate, ETC-1200-CoA, by an endogenous liver very long-chain acyl-CoA synthetase-1 (ACSVL1). Since ACSVL1 is not present in the skeletal muscle, less risk of myalgia symptoms and myopathy is expected. In a Mendelian randomization study, combined exposure to variants in the ACLY, HMGCR and Niemann-Pick C1-Like 1 genes produced an additive decrease in low-density lipoprotein cholesterol levels and additive reduction in the risk of cardiovascular events. The aim of this review is to describe bempedoic acid mechanism of action, and its pharmacokinetic and pharmacodynamic characteristics.

Bempedoic acid: effects on lipoprotein metabolism and atherosclerosis.

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Bempedoic acid has emerged as a potent inhibitor of ATP-citrate lyase (ACLY), a target for the reduction of LDL cholesterol (LDL-C). We review the impact of bempedoic acid treatment on lipoprotein metabolism and atherosclerosis in preclinical models and patients with hypercholesterolemia. The liver-specific activation of bempedoic acid inhibits ACLY, a key enzyme linking glucose catabolism to lipogenesis by catalyzing the formation of acetyl-CoA from mitochondrial-derived citrate for de novo synthesis of fatty acids and cholesterol. Adenosine monophosphate-activated protein kinase activation by bempedoic acid is not required for its lipid-regulating effects in vivo. Mendelian randomization of large human study cohorts has validated ACLY inhibition as a target for LDL-C lowering and atheroprotection. In rodents, bempedoic acid decreases plasma cholesterol and triglycerides, and prevents hepatic steatosis. In apolipoprotein E-deficient (Apoe) mice, LDL receptor-deficient (Ldlr) mice and LDLR-deficient miniature pigs, bempedoic acid reduces LDL-C and attenuates atherosclerosis. LDLR expression and activity are increased in primary human hepatocytes and in Apoe mouse liver treated with bempedoic acid suggesting a mechanism for LDL-C lowering, although additional pathways are likely involved. Phase 2

and 3 clinical trials revealed that bempedoic acid effectively lowers LDL-C as monotherapy, combined with ezetimibe, added to statin therapy and in statin-intolerant hypercholesterolemic patients. Treatment does not affect plasma concentrations of triglyceride or other lipoproteins. The LDL-C-lowering and attenuated atherosclerosis in animal models and reduced LDL-C in hypercholesterolemic patients has validated ACLY inhibition as a therapeutic strategy. Positive results from phase 3 long-term cardiovascular outcome trials in high-risk patients are required for bempedoic acid to be approved for prevention of atherosclerosis.

Phase 1, Single- and Multiple-Ascending-Dose, Food-Effect, and East Asian Subject Studies to Assess the Pharmacokinetics, Safety, and Tolerability of Bempedoic Acid, a Selective Inhibitor of Adenosine Triphosphate Citrate Lyase.

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Bempedoic acid is an adenosine triphosphate citrate lyase inhibitor that lowers low-density lipoprotein cholesterol by inhibiting cholesterol synthesis and upregulating hepatic low-density lipoprotein receptor expression. After oral dosing, bempedoic acid was readily absorbed, attaining maximum concentrations with a median time of 3.5 hours, and may be taken without regard to food. Steady-state oral pharmacokinetics in healthy adults receiving bempedoic acid at the approved 180 mg/day dose were characterized by mean maximum concentration of 20.6 µg/mL, area under the concentration-time curve over 24 hours of 289 µg·h/mL, and elimination half-life of 21.1 hours. Multiple-dose pharmacokinetics were linear at bempedoic acid doses of 120-220 mg/day. Circulating concentrations of the active metabolite ESP15228 were 18.0% of bempedoic acid concentrations on average. Comparisons of bempedoic acid 180 mg/day pharmacokinetics after single and multiple dosing revealed no clinically meaningful differences between Japanese, Chinese, and Western subjects. Mean estimates of bempedoic acid elimination half-life in Japanese (25.2 hours) and Chinese (20.0 hours) subjects were comparable to Western subjects (23.9 hours) following 14 days of once-daily dosing. Bempedoic acid was generally safe and well tolerated up to a dose of 220 mg/day across the study populations described herein.

Role of Bempedoic Acid in Clinical Practice.

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Many patients do not achieve optimal low-density lipoprotein cholesterol (LDL-C) levels with statins alone; others are unable to tolerate statin therapy. Additional non-statin treatment options including ezetimibe, proprotein convertase subtilisin/kexin type 9 inhibitors, and bile acid sequestrants are often necessary to further reduce the risk of atherosclerotic cardiovascular disease. This review provides practical guidance as to the use of bempedoic acid to lower LDL-C and includes direction as to which patients may benefit and advice for safety monitoring during treatment. Bempedoic acid, a new class of agent, is a prodrug converted to bempedoyl-CoA by very long-chain acyl-CoA synthetase 1, an enzyme with high expression in the liver but that is undetectable in the skeletal muscle. Bempedoic acid inhibits the enzyme adenosine triphosphate (ATP)-citrate lyase, which lies two steps upstream from β-hydroxy β-methylglutaryl-CoA reductase in the cholesterol biosynthesis pathway. In clinical trials conducted in patients with or at risk for atherosclerotic cardiovascular disease or familial heterozygous hypercholesterolemia, bempedoic acid in combination with statins and/or ezetimibe significantly reduced

LDL-C, apolipoprotein B, and high-sensitivity C-reactive protein compared with placebo. Bempedoic acid is generally well tolerated with no clinically meaningful increase in muscle-related symptoms relative to placebo, even in patients taking maximally tolerated statins. A small increase in serum uric acid (mean increase 0.8 mg/dL) is the most noteworthy adverse effect. Bempedoic acid provides an effective and generally well-tolerated medication to further reduce LDL-C in patients taking maximally tolerated statins or manage LDL-C levels in those who are unable to take statins. The potential for a reduced incidence of major cardiovascular events with bempedoic acid is being investigated in the CLEAR Outcomes trial, with results expected in 2023.
